

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY



DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

ASSIGNMENT 1, MICROPROCESSOR AND INTERFACING DECEMBER 2024

(10 marks)

1. 8085 microprocessor Registers

- i) Describe the function of program counter in 8085 microprocessor?
- ii) Discuss the use of stack pointer
- iii) Discover why data bus is bi-directional?

2. Instruction execution and Timing diagram

- i) Explain the Timing diagram of STA 526A
- ii) Evaluate the timing diagram of OUT instruction.
- iii) Illustrate the timing diagram for LXI H, 4200H

3. 8085 microprocessor interrupts

- i) Classify the types of interrupts in 8085. Explain in detail about the hardware interrupts in 8085.
- ii) Summarize the function of trap interrupt and its significance.
- iii) Explain the level triggered interrupt. Which of the interrupts in 8085 are level triggered?

4. 8085 microprocessor Assembly language programming

- i) Demonstrate an 8085 program to swap lower and higher nibble of the contents of accumulator
- ii) Discuss the function of CALL instruction *with example*
- iii) Assess the time delay generated using subroutines. *demonstrate with example*
- iv) Write an assembly language program to find two's complement of a 16 bit data.
- v) Write the assembly language finding the biggest number in a block of data stored in the memory locations from 4270H – 427FH.
- vi) Develop an assembly language program for the loop structure with counting of 10 numbers.
- vii) Write an 8085 assembly language program to divide a 8 bit number by another 8 bit number and store the remainder and quotient in memory locations 4252 and 4253 respectively.
- viii) Write an 8085 assembly language program to solve the following equation: $Z=2X+Y$ where X and Y are stored in memory location 4200 and 4201 respectively. The value of Z should be stored in 4202 (Lower byte) and 4203(higher byte).

Group assignment, submit on or before 12/12/2024 Thursday